

Acemoglu & Restrepo (2018)

The Race between Man and Machine — displacement, reinstatement, and the condition under which automation raises wages

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Repository 6 · Week of September 21 – 26, 2026

github.com/Arriola123456/ai-06-restrepo

Version read: NBER Working Paper 22252 as revised in June 2017 (87 pp., the course PDF; title *The Race Between Machine and Man*), pinned by SHA-256 in `lean/`. Page numbers are the PDF's (printed page = PDF page – 2). Backup slides after slide 14.

1 · The paper and the problem: the unit of analysis is the economy

Question. Technology enters as *automation* (machines take over tasks previously done by labor) and as *new tasks* (labor gets new things to do). What does each do to output, wages, the labor share and employment — and can the economy keep a stable labor share when both run at once?

This is the only paper of the term whose agent is the aggregate economy. There is no individual optimising over a menu of AI tools: there is a unit measure of tasks $i \in [N - 1, N]$, competitive firms that pick the cheaper factor for each task, a representative household supplying labor, and a capital stock K . The “problem” is a competitive equilibrium, and the objects that move are I (how far automation has reached) and N (how far the task frontier has moved).

Environment (Section 2.1, p. 8–9). Final good $Y = \tilde{B} \left(\int_{N-1}^N y(i)^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}}$. Tasks $i \leq I$ are technologically automated: capital and labor are perfect substitutes there, $k(i) + \gamma(i)l(i)$; tasks $i > I$ need labor with productivity $\gamma(i)$. **Assumption 1:** γ strictly increasing — labor’s comparative advantage rises with the task index. Preferences $u(C, L)$ with an increasing labor supply $L = L^s(\omega)$, $\omega = W/(RK)$ (eq. 11). **Assumption 2:** homothetic demand ($\eta \rightarrow 0$ or $\zeta = 1$), so $\hat{\sigma} = \sigma(1 - \eta) + \zeta\eta$ is the elasticity between capital and labor. **Assumption 3:** $K < \underline{K}$, i.e. $R > W/\gamma(N)$: new tasks are adopted at once.

1 · The threshold: who does which task (p. 10)

Tasks are priced at minimum unit cost,

$$p(i) = \begin{cases} \min\{R, W/\gamma(i)\}^{1-\eta} & i \leq I \\ (W/\gamma(i))^{1-\eta} & i > I. \end{cases}$$

Since γ is increasing there is a unique $\tilde{I}$ with $\frac{W}{R} = \gamma(\tilde{I})$ (eq. 6): capital is cheaper below it, labor above it. Technology may bind before cost does, so the equilibrium threshold is

$$I^* = \min\{I, \tilde{I}\} : \quad i < I^* \text{ produced with capital,} \quad i > I^* \text{ with labor.}$$

Two regimes run through everything that follows:

- ▶ **technology-constrained**, $I^* = I < \tilde{I}$: firms would automate more if they could;
- ▶ **cost-minimising**, $I^* = \tilde{I} < I$: the frontier is slack; factor prices decide.

Proposition 1 (p. 12). Unique equilibrium; output is a CES aggregate with endogenous shares,

$$Y = \frac{B}{1-\eta} \left[(I^* - N + 1)^{\frac{1}{\sigma}} K^{\frac{\hat{\sigma}-1}{\hat{\sigma}}} + \left(\int_{I^*}^N \gamma(i)^{\hat{\sigma}-1} di \right)^{\frac{1}{\hat{\sigma}}} L^{\frac{\hat{\sigma}-1}{\hat{\sigma}}} \right]^{\frac{\hat{\sigma}}{\hat{\sigma}-1}},$$

and ω solves the relative demand (13). **Corollary 1:** with $\sigma = \zeta = 1$, $\gamma \equiv 1$, $Y = \frac{B}{1-\eta} K^{1-N+I^*} L^{N-I^*}$: the labor share is $N - I^*$.

2 · The main result I: Proposition 2, the two forces are polar (p. 11)

Let $\varepsilon_L > 0$ be the labor-supply elasticity and $\Lambda_I = \frac{\gamma(I^*)^{\hat{\sigma}-1}}{\int_{I^*}^N \gamma^{\hat{\sigma}-1}} + \frac{1}{I^* - N + 1} > 0$,

$$\Lambda_N = \frac{\gamma(N)^{\hat{\sigma}-1}}{\int_{I^*}^N \gamma^{\hat{\sigma}-1}} + \frac{1}{I^* - N + 1} > 0.$$

Technology-constrained case ($I^* = I < \tilde{I}$):

$$\frac{d \ln(W/R)}{dI} = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} < 0, \quad \frac{d \ln(W/R)}{dN} = \frac{\Lambda_N}{\hat{\sigma} + \varepsilon_L} > 0, \quad \frac{d \ln(W/R)}{d \ln K} = \frac{1 + \varepsilon_L}{\hat{\sigma} + \varepsilon_L} > 0.$$

Cost-minimising case ($I^* = \tilde{I} < I$): $d \ln(W/R)/dI = 0$ and $d \ln(W/R)/dN = \Lambda_N/(\sigma_{free} + \varepsilon_L) > 0$ with $\sigma_{free} = \hat{\sigma} + \Lambda_I/\varepsilon_\gamma > \hat{\sigma}$.

In all cases the labor share and employment move with ω : $dL/dN > 0$, and when $I^* = I$, $dL/dI < 0$. So automation is always capital-biased (lowers W/R , s_L , L) and new tasks are always labor-biased (raise all three) — *displacement* against *reinstatement*.

2 · The main result II: Proposition 3, the wage is the ambiguous object (p. 13)

Constrained case, with $W/\gamma(I^*) > R > W/\gamma(N)$:

$$d \ln Y|_{K,L} = \underbrace{\frac{B^{\hat{\sigma}-1}}{1-\hat{\sigma}} \left(\left(\frac{W}{\gamma(I^*)} \right)^{1-\hat{\sigma}} - R^{1-\hat{\sigma}} \right)}_{\text{productivity effect of automation} > 0} dI + \frac{B^{\hat{\sigma}-1}}{1-\hat{\sigma}} \left(R^{1-\hat{\sigma}} - \left(\frac{W}{\gamma(N)} \right)^{1-\hat{\sigma}} \right) dN,$$

$$d \ln W = d \ln Y|_{K,L} + (1-s_L) \left(\frac{\Lambda_N}{\hat{\sigma}+\varepsilon_L} dN - \underbrace{\frac{\Lambda_I}{\hat{\sigma}+\varepsilon_L} dI}_{\text{displacement}} \right), \quad d \ln R = d \ln Y|_{K,L} - s_L \left(\frac{\Lambda_N}{\hat{\sigma}+\varepsilon_L} dN - \frac{\Lambda_I}{\hat{\sigma}+\varepsilon_L} dI \right).$$

New tasks always raise W ; automation always raises R and **raises W iff the productivity effect exceeds the displacement effect**. The productivity effect is the cost saving $W/\gamma(I^*) - R$ on the marginal automated task; the displacement effect is $(1-s_L)\Lambda_I/(\hat{\sigma}+\varepsilon_L)$. Cost-minimising case: automation does nothing to factor prices; new tasks raise W and may lower R .

Conditions, spelled out. Assumptions 1–3; $\hat{\sigma} \neq 1$ in the displayed productivity term (Cobb-Douglas is the limit); K fixed (static); the sign of $d \ln W/dI$ depends on K — next slide.

2 · The trap: “does automation necessarily reduce wages and the labour share?”

	$I \uparrow$, constrained	$I \uparrow$, cost-min.	$N \uparrow$
$W/R, s_L, L$	fall, always (Prop. 2)	unchanged	rise, always
R	rises, always (Prop. 3)	unchanged	may fall
W	ambiguous : productivity vs displacement	unchanged	rises, always
$Y _{K,L}$	rises	unchanged	rises

Flat answer: “yes, automation lowers wages and the labor share.” **Paper:** the labor share, yes, always (when technology binds); the wage, no — it rises when the cost saving on the marginal task is large.

The condition, in closed form (Corollary 1, my derivation, proved in Lean): with $Y = cK^{1-s}L^s$, $s = N - I^*$, $\ln W = \ln c + \ln s + (1 - s) \ln K + (s - 1) \ln L$, so

$$\frac{d \ln W}{d I^*} = \ln \frac{K}{L} - \frac{1}{N - I^*} > 0 \iff \frac{K}{L} > e^{1/(N - I^*)} =: \bar{K}.$$

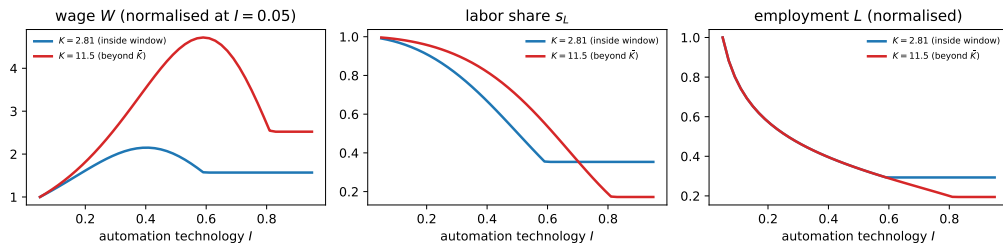
Automation raises the wage when capital is *abundant* enough — R low, the saving $W - R$ large. And $\bar{K}/L = e^{1/s} > \underline{K}/L = (1 - s)/s$: inside Assumption 3 ($K < \underline{K}$) the wage always falls. **Proposition 3 prints the opposite direction** (“increases the wage when $K < \bar{K}$ ”); slide 12.

3 · What I did analytically and computationally

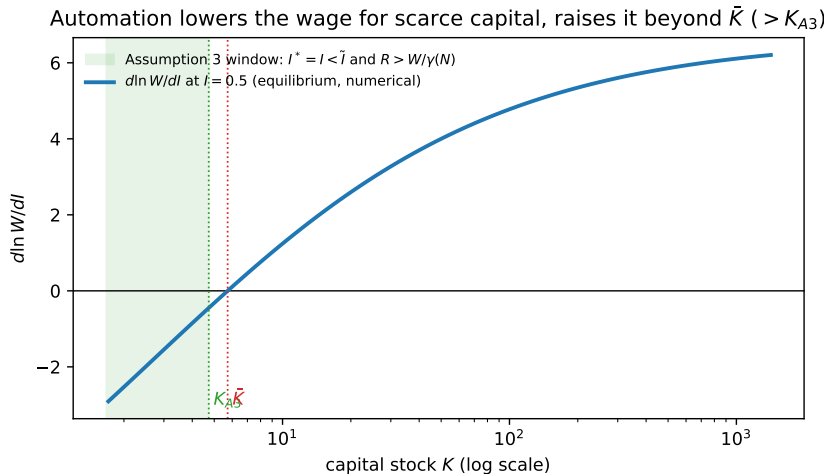
- ▶ **By hand** (hand/, slide 13): the cost threshold and I^* ; the Cobb-Douglas case — W , R , $s_L = N - I^*$ and the sign of $d \ln W / dI^*$.
- ▶ **Numerics** (analysis/task_model.py): the static equilibrium of Section 2 solved from eq. (13) with $\gamma(i) = 0.3e^{3i}$, $\hat{\sigma} = 0.6$, $\varepsilon_L = 0.3$, $L^s(\omega) = \omega^{\varepsilon_L}$; the two regimes detected from $W/R \gtrless \gamma(I)$; prices from the ideal price index (10).
- ▶ **Check against the paper**: the numerical total derivative $d \ln W / dI$ equals Proposition 3's formula (productivity minus displacement) to four decimals at every point tried.
- ▶ **Findings**: s_L and L fall with I everywhere; W falls throughout the Assumption-3 window ($K < \underline{K} = 4.73$ at $I = 0.5$) and rises only beyond $\bar{K} = 5.73 > \underline{K}$; new tasks raise W and s_L ; the closed-form Cobb-Douglas threshold $\bar{K}/L = e^{1/(N-I^*)}$ matches.
- ▶ **Lean via AppliedModelingLib** (slides 10–12): ten source-facing statements of Section 2 proved; the Cobb-Douglas wage response as an extension. Status partially formalized.

3 · Automation: s_L and L always fall, W may go either way

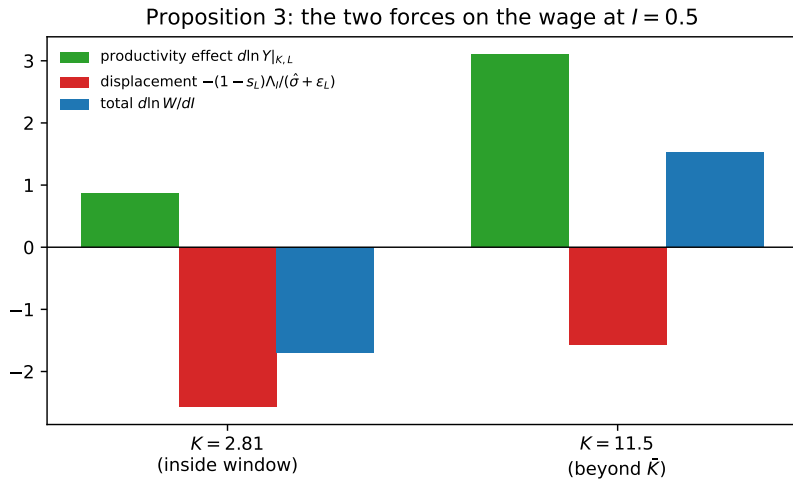
Automation: s_L and L always fall; W falls inside the window, rises beyond $\bar{K}$ ($\hat{\sigma}=0.6$, $\varepsilon_L=0.3$)



3 · $d \ln W / dl$ against K : negative inside Assumption 3, positive beyond $\bar{K}$



3 · Proposition 3 decomposed: productivity effect against displacement effect



4 · Lean I: the run, the statement spec, the status

Workflow (AppliedModelingLib, clone at 2db7d108). Pin the June-2017 PDF by SHA-256 → `init-spec` → statement spec with 10 targets (PDF-page locators, literal source statements, exact Lean types) → `paper_contribution.py new -statement-spec` (the scaffold Lean-validates every Spec) → proofs in `ProofInterface.lean`, extension in `MainTheorems.lean` → `lake build AR18RaceManMachine` → `check AR18RaceManMachine -fast`. **Agent:** Claude Code, not Codex (my decision, four hours before the deadline; disclosed in `lean/docs/RUN_LOG.md`).

The ten Specs (`lean/PaperInterface.lean`), all proved, no sorry: the cost threshold of (5)–(6) with γ strictly increasing; $I^* = \min\{I, \tilde{I}\}$; $\Lambda_I, \Lambda_N > 0$; Prop. 2's three signs in the constrained case and $\sigma_{free} > \hat{\sigma}$; Prop. 3's productivity effect > 0 on both sides of $\hat{\sigma} = 1$; automation raises R ; the wage decomposition as an iff; new tasks raise W ; Corollary 1's marginal products as real derivatives and the labor share $N - I^*$.

Result: `lake build`: Build completed successfully (3313 jobs); `check -fast`: exit 0. **Status: partially formalized.** The equilibrium map (13) and the derivatives of Prop. 2 are taken as the paper's displays; the general $\bar{K}$ is not derived. **Blocker, stated exactly:** the library root `AppliedModelingLib.olean` does not build on my laptop (disk-bound, 12 h last week), so the paper modules import three `Mathlib` modules instead of `import Mathlib`, and the spec validation ran under those imports instead of `import AppliedModelingLib`. No library declaration is used.

4 · Lean II (required slide): the cost threshold of equations (5)–(6)

Paper (p. 10). $p(i) = \min\{R, W/\gamma(i)\}^{1-\eta}$ for $i \leq I$; “there is a unique threshold $\tilde{I}$ such that $W/R = \gamma(\tilde{I})$. For all tasks $i < \tilde{I}$, we have $R < W/\gamma(i)$ ” (Assumption 1: γ strictly increasing).

```
def paper_cost_thresholdSpec : Prop :=
  ∀ (W R : ℝ) (γ : ℝ → ℝ) (i It : ℝ),
    0 < W → 0 < R → StrictMono γ → (∀ x, 0 < γ x) → γ It = W / R →
      (i < It → R < W / γ i) ∧ (It < i → W / γ i < R)

theorem paper_cost_threshold : paper_cost_thresholdSpec := by
  intro W R γ i It hW hR hmono hpos hIt
  have hγi : 0 < γ i := hpos i
  constructor
  · intro hi
    have h1 : γ i < γ It := hmono hi          -- Assumption 1 does the work
    rw [hIt] at h1                            -- γ i < W / R
    rw [lt_div_iff0 hR] at h1                 -- γ i * R < W
    rw [lt_div_iff0 hγi]                     -- goal: R * γ i < W
    linarith [mul_comm R (γ i)]
  · intro hi
    have h1 : γ It < γ i := hmono hi
    rw [hIt] at h1; rw [div_lt_iff0 hR] at h1; rw [div_lt_iff0 hγi]
    linarith [mul_comm R (γ i)]
```

Reading. W, R reals, γ a real function, $\tilde{I}$ (It) any real with $\gamma(\tilde{I}) = W/R$; StrictMono γ is Assumption 1 verbatim. The nontrivial step is `lt_div_iff0`, which moves the positive denominator across the inequality; then the claim is linear. Lean verifies the sentence *exactly*, with one hypothesis the paper leaves implicit: $\gamma > 0$. Uniqueness of $\tilde{I}$ is not needed by the row. lake build: success; check -fast: exit 0.

4 · Lean III: the Cobb-Douglas wage response — the extension that settles the trap

Claim (mine). With $\ln W = \ln c + \ln s + (1 - s) \ln K + (s - 1) \ln L$, $s = N - I^*$:

$d \ln W / ds = 1/s - \ln K + \ln L$, hence $d \ln W / dI^* = \ln(K/L) - 1/(N - I^*) > 0 \iff e^{1/(N - I^*)} < K/L$.

```
theorem cobb_douglas_log_wage_deriv (c K L s : ℝ) (hs : 0 < s) :
  HasDerivAt (fun s' : ℝ => Real.log c + Real.log s' + (1 - s') * Real.log K + (s' - 1) * Real.log L)
    (1 / s - Real.log K + Real.log L) s := by
  have h1 : HasDerivAt (fun s' : ℝ => Real.log s') (1 / s) s := by
    simp [one_div] using Real.hasDerivAt_log hs.ne'
  have h2 : HasDerivAt (fun s' : ℝ => (1 - s') * Real.log K) (-Real.log K) s := by
    have := ((hasDerivAt_id' (x := s)).const_sub 1).mul_const (Real.log K); simp using this
  have h3 : HasDerivAt (fun s' : ℝ => (s' - 1) * Real.log L) (Real.log L) s := by
    have := ((hasDerivAt_id' (x := s)).sub_const 1).mul_const (Real.log L); simp using this
  have h := (((hasDerivAt_const s (Real.log c)).add h1).add h2).add h3
  refine h.congr_deriv ?_; ring

theorem cobb_douglas_wage_rises_iff (K L s : ℝ) (hK : 0 < K) (hL : 0 < L) (hs : 0 < s) :
  0 < Real.log (K / L) - 1 / s ↔ Real.exp (1 / s) < K / L := by
  have hKL : 0 < K / L := div_pos hK hL
  rw [sub_pos]; exact Real.lt_log_iff_exp_lt hKL
```

Reading. The derivative is assembled term by term with `HasDerivAt` combinators (`Real.hasDerivAt_log`

needs only $s \neq 0$); `congr_deriv` plus `ring` matches the paper's form. The sign lemma is

`Real.lt_log_iff_exp_lt`. Since $s = N - I^*$ and $dI^* = -ds$, the wage rises with automation iff

$K/L > e^{1/(N - I^*)}$: **abundant capital**. Under Assumption 3, $K/L < (1 - s)/s < e^{1/s}$, so the printed “increases the wage when $K < \bar{K}$ ” cannot be right for this case; the numerics agree for $\hat{\sigma} \neq 1$.

5 · Where I did not believe the AI — and where I did not believe the paper

hand derivation
hand/

Hand derivation: the threshold
and the Cobb-Douglas wage. Full
pages in the backup.

Claim 1 (the flat answer): “automation lowers wages and the labor share.” Props. 2–3: the labor share always falls when technology binds; the wage rises iff productivity beats displacement, and does not move in the cost-minimising regime. **Verdict: half right.**

Claim 2 (my first numerics): I wrote $W/R = \omega K/L$; Prop. 2’s $d \ln(W/R)/dI = d \ln \omega/dI$ then failed by $(1 - \varepsilon_L)$. The paper’s $\omega = W/(RK)$ gives $W/R = \omega K$; fixed, the numerical derivative matches Prop. 3’s formula to four decimals. **Verdict: I was wrong; the check caught it.**

Claim 3 (the paper): “there exists $\bar{K} > \underline{K}$ such that an increase in I increases the wage when $K < \bar{K}$ and reduces it when $K > \bar{K}$.” By hand and in Lean, Cobb-Douglas gives $d \ln W/dI^* = \ln(K/L) - 1/(N - I^*)$: the wage rises for *large* K , so with $\bar{K} > \underline{K}$ it always falls inside Assumption 3; the $\hat{\sigma} = 0.6$ numerics and the paper’s own intuition (cost savings $W/\gamma(I^*) - R$) agree. **Verdict: the two inequalities are printed the wrong way round** (AER text not checked).

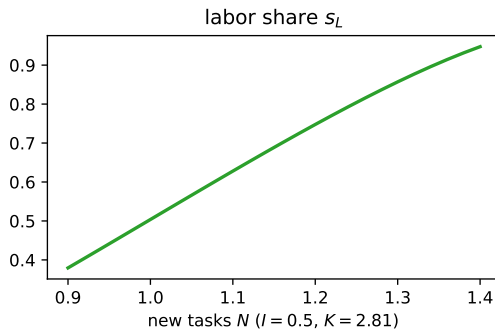
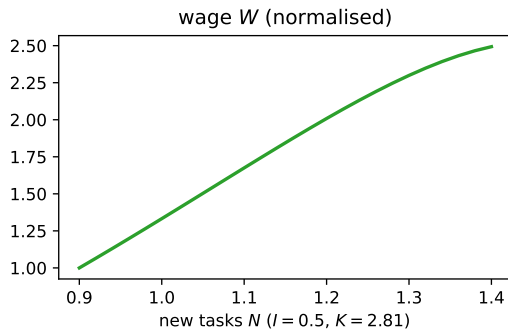
Claim 4 (Lean pushed back): the productivity term divides by $1 - \hat{\sigma}$; the sign proof needs $\hat{\sigma} < 1$ and $\hat{\sigma} > 1$ separately and excludes $\hat{\sigma} = 1$ — the Cobb-Douglas limit the paper takes silently.

5 · What to carry away

- ▶ **Two forces, not one.** Automation *displaces* (fewer tasks for labor: W/R , s_L , L fall) but also raises productivity; new tasks *reinstates* labor. The race is between I and N .
- ▶ **The wage condition.** Automation raises the wage iff the cost saving on the marginal task, $W/\gamma(I^*) - R$, outweighs $(1 - s_L)\Lambda_I/(\hat{\sigma} + \varepsilon_L)$ — i.e. when capital is abundant (Cobb-Douglas: $K/L > e^{1/(N-I^*)}$). Under the paper's Assumption 3 it falls.
- ▶ **Regime matters.** If technology does not bind ($I^* = \tilde{I} < I$), better automation technology changes nothing; only N moves prices.
- ▶ **Lean.** Ten rows proved exactly from the displays; the equilibrium map and the general $\bar{K}$ are the declared boundary; the extension proves the closed-form threshold and exposes the reversed inequality.
- ▶ **Versions.** NBER June 2017 (course PDF) read and pinned; AER 2018 reverses the title; the page numbers here are the PDF's.

B1 · New tasks: W and s_L always rise (reinstatement)

New tasks (reinstatement): wage and labor share always rise (Propositions 2-3)



hand derivation (hand/)

B3 · Lean IV: Proposition 3's productivity effect on both sides of $\hat{\sigma} = 1$

```
def paper_prop3_productivity_effect_positiveSpec : Prop :=
  ∀ Wg R σh : ℝ, 0 < R → R < Wg → σh ≠ 1 →
    0 < (Wg ^ (1 - σh) - R ^ (1 - σh)) / (1 - σh)

theorem paper_prop3_productivity_effect_positive : paper_prop3_productivity_effect_positiveSpec := by
  intro Wg R σh hR hlt hne
  rcases lt_or_gt_of_ne hne with hlt1 | hgt1
  · have hexp : 0 < 1 - σh := by linarith
    have h : R ^ (1 - σh) < Wg ^ (1 - σh) := Real.rpow_lt_rpow hR.le hlt hexp
    exact div_pos (by linarith) hexp
  · have hexp : 1 - σh < 0 := by linarith
    have h : Wg ^ (1 - σh) < R ^ (1 - σh) := Real.rpow_lt_rpow_of_neg hR hlt hexp
    exact div_pos_of_neg_of_neg (by linarith) hexp

def paper_prop3_wage_decompositionSpec : Prop :=
  ∀ P sL ΔI σh εL dI : ℝ,
    0 < P - (1 - sL) * ((1 / (σh + εL)) * ΔI * dI) ↔
    (1 - sL) * ((1 / (σh + εL)) * ΔI * dI) < P
```

Reading. Wg is $W/\gamma(I^*)$ and R the rental rate; the hypothesis $R < Wg$ is the constrained case of Proposition 3. For $\hat{\sigma} < 1$ the real power is increasing and both numerator and denominator are positive; for $\hat{\sigma} > 1$ both are negative. The decomposition row states the wage sign as an equivalence over the paper's display: productivity effect P against displacement.